SIMATS ENGINEERING

ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 15

Duration : 60 Minutes

Code of Conduct

I **V.PREETE(192572004)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.PREETE

Assessment Tasks

MCQ Test

1.A company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?

a) 5 TB
b) 10 TB
c) 15 TB
d) 20 TB

ANS: c) 15 TB

2. A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

a) ~4 minutes
b) ~40 minutes
c) ~4 hours
d) ~24 hours

ANS: a) ~4 minutes

3.A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is the average number of requests per server per second?

a)5000
b) 6000
c) 7000
d) 10,000

ANS: b) 6000

4. A company's cloud network has 1 Gbps bandwidth and needs to transfer 500 GB of data. How long will it take?

1 hour
b) 10 hours
c) 1.1 hours
d) 5.5 hours

ANS: c) 1.1 hours

5. A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

a) ~4 hours
b) ~8 hours
c) ~20 hours
d) ~40 hours

ANS: a) ~4 hours

6. A cloud provider offers pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10 TB, what is the monthly cost?

a) $50
b) $500
c) $1000
d) $2000

ANS: b) $500

7. A cloud network has 10 Gbps throughput but experiences 2% packet loss. What is the effective throughput?

a) 9.8 Gbps
b) 9.6 Gbps
c) 8.0 Gbps
d) 7.5 Gbps

ANS: a) 9.8 Gbps

8.A user uploads 250 MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

a) 91.25 GB
b) 85 GB
c)100 GB
d) 120 GB

ANS: a) 91.25 GB

9.A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

a) 500 Mbps
b) 250 Mbps

c) 200 Mbps
d) 1000 Mbps

ANS: a) 500 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

a) 262 minutes
b) 365 minutes
c) 525 minutes
d) 150 minutes

ANS: a)262 minutes

11. A virtualized data center has 200 physical servers. If each server runs 40 virtual machines, what is the total number of VMs in the data center?

a) 4,000
b) 8,000
c) 6,000
d) 10,000

ANS: b) 8,000

12.A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?

a) 4 GB
b) 6 GB
c) -8 GB
d) 10 GB

ANS: a) 4 GB

13.A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an overcommitment ratio of 2:1, how many VMs can be hosted?

a)16
b) 32
c) 64
d) 128

ANS: b) 32

14.A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?

a) 40 kW
b) 50 kW
c) 60 kW
d) 70 kW

ANS: a) 40 kW

15.A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

a) 6,400
b) 12,800
c) 25,600
d) 50,000

ANS: c) 25,600

16.Which cloud deployment model is used for organizations that require both private and public cloud resources?

A) Hybrid Cloud
B) Public Cloud
C) Private Cloud
D) Community Cloud

ANS: A) Hybrid Cloud

17.Which cloud security strategy ensures that sensitive data remains private even if intercepted?

A) Encryption
B) Load balancing
C) Virtualization
D) Resource pooling

ANS: A) Encryption

18. If a hypervisor hosts 50 VMs and each VM uses 20 GB bandwidth per day, calculate total bandwidth consumption per week.

a)5000 GB
b) 7000 GB
c) 6000 GB
d) 7500 GB

ANS: b) 7000 GB

19.A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

a) 56 GHz
b) 58 GHz
c) 64 GHz
d) 70 GHz

ANS: a) 56 GHz

20. If the network latency in a virtualized system is 5 ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

a) 1.5 sec
b) 2 sec
c) 1 sec
d) 3 sec

ANS: a) 1.5 sec
21.If 100 VMs transfer logs of 200 MB/day each to the cloud, calculate the total data transferred in a 30-day month.

a) 600 GB
b) 500 GB
c) 700 GB
d)800 GB

ANS: a) 600 GB

22. A cloud service's availability is 99.95%. How much downtime (in minutes) is expected in a month (30 days)?

a) 21.6 min
b) 36 min
c) 43.2 min

d) 18 min

ANS: a) 21.6 min

23. If a cloud server handles 200 requests/sec and each request uses 10 KB memory buffer, compute memory usage per minute.

a) 120 MB
b) 100 MB
c) 150 MB
d) 200 MB

ANS: a) 120 MB

24. A hypervisor hosts 120 VMs. If 75% of them require 2 vCPUs each, calculate total vCPU requirement.
a) 180
b) 150
c) 120
d) 160

ANS: a) 180

25.A company has 20 racks with 20 servers each. If each server uses 3 virtual NICs, how many vNICs exist in total?
a) 1200
b) 1000
c) 800
d) 60

ANS: a) 1200

26. If a hypervisor can boot a VM in 2 minutes, how long will it take to boot 60 VMs in batches of 10 concurrently?
a) 12 min
b) 10 min
c) 6 min
d) 8 min

ANS: a) 12 min
27. A cloud data center uses 5 cooling units, each consuming 3.5 kW. Find total cooling energy per day.
Formula:
Energy = Units × Power × Time (hrs)
a) 420 kWh
b) 480 kWh

c) 400 kWh
d) 500 kWh

ANS: a) 420 kWh

28. Each server is configured with 2 TB storage. A virtual SAN clusters 20 such servers. What is total usable capacity, assuming 20% is used for redundancy?
a) 32 TB
b) 35 TB
c) 30 TB
d) 28 TB

ANS: a) 32 TB

***29.*Which of the following is NOT a benefit of virtualization?**

a) Reduced hardware costs
b) Increased energy consumption
c) Improved resource management
d) Faster deployment of applications

ANS: b) Increased energy consumption

***30.*Which feature in virtualization allows multiple OS environments to run simultaneously?**

a) Hypervisor
b) Load balancer
c) Firewall
d) BIOS

ANS: a) Hypervisor